

E-commerce Cohort Analysis Report: 2024 - 2025

Source: This analysis is created by a databricks notebook “src_orders_2024_2025” and dashboard “cohort_customers 2026-01-27”

Data Pipeline Architecture: The Medallion Framework

To ensure data integrity and scalable analysis across the 24-month period, the pipeline follows the **Medallion Architecture**. This modular approach allows for a clear separation between raw data collection and final business insights.

1. Bronze Layer (Raw Data Ingestion)

- **Source:** Direct ingestion of the `ecom_orders_2024_rebalanced.csv` and `ecom_orders_incremental_2025_rebalanced.csv` datasets.
- **State:** Contains raw, unprocessed transaction logs.
- **Audit Note:** This layer was used to identify the data anomaly on **December 31, 2024**, which has been flagged for further verification.

2. Silver Layer (Data Cleansing & Enrichment)

- **Transformation:** This is the "Clean Room" where data is deduplicated and standardized.
- **Key Logic:** The **First-Purchase Date** is calculated at this stage for every unique Customer ID. This is a critical step that creates the "Source of Truth" for cohort assignment.
- **Stability:** By fixing the start date in the Silver Layer, we ensure that customers remain tied to their original cohorts as the pipeline updates incrementally.
- **Output:** Calculation of final KPI **Repeat Purchase Depth%** (2+, 3+, 4+ purchases).
- **Application:** This layer fuels the 2 charts presented in this report, providing the basis for our strategic business recommendations.

3. Gold Layer (Business Intelligence & Insights)

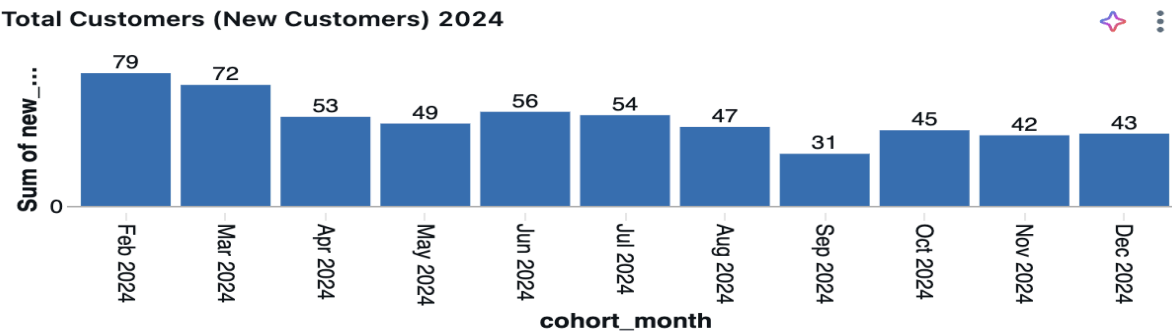
- **Aggregation:** Data is modeled specifically for the Cohort Analysis dashboard.
- **Output:** Calculation of final KPIs, including **Retention %** and **Total customers (new customers)**.
- **Application:** This layer fuels the 6 charts presented in this report, providing the basis for our strategic business recommendations.

Why this matters for the Business:

- **Traceability:** We can trace any metric in the Gold Layer back to the raw Bronze records.
- **Reliability:** Technical anomalies (like the Dec 31st gap) are isolated in the Silver Layer, ensuring they do not distort the high-level trends shown to stakeholders.
- **Scalability:** The pipeline is designed to handle future incremental updates (e.g., 2026 data) without rebuilding the underlying logic.

1. 2024 Performance Overview (Initial Baseline)

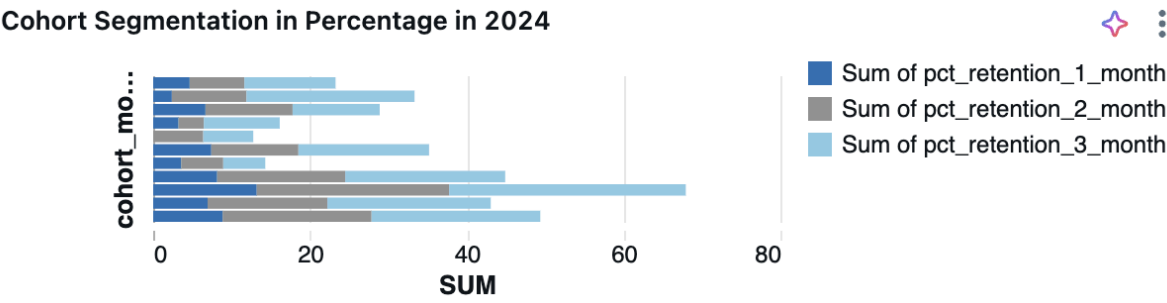
Cohort Size 2024



Referencing: "Total Customers (New Customers) 2024" Chart

- **Acquisition Trend:** The year started strong with a peak of 79 new customers in February 2024. However, acquisition saw a steady decline throughout the year, hitting a low of 31 in September before stabilizing around 43 customers per month in Q4.

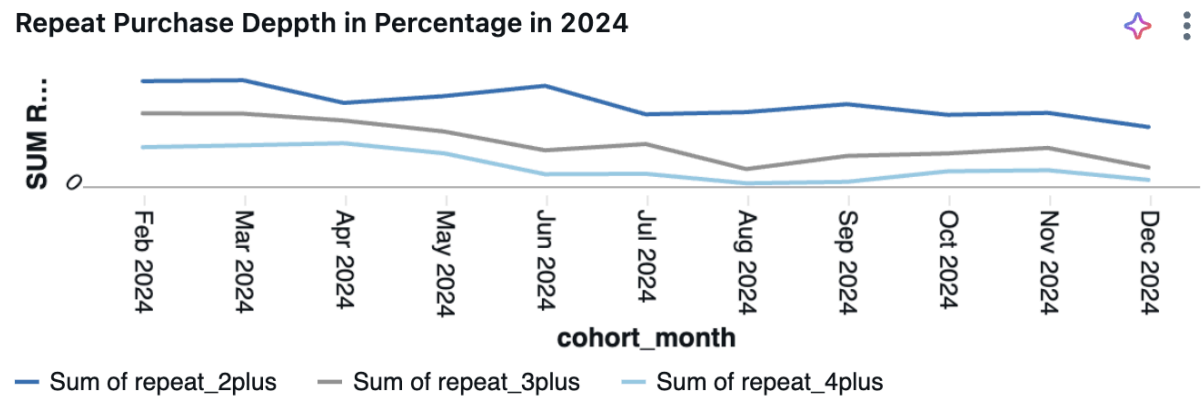
Retention to 2nd Purchase 2024



Referencing: "Cohort Segmentation in Percentage 2024" Chart

- Retention Trend:** The 1-month retention (`pct_retention_1_month`) remained relatively low but stable. April 2024 stands out as a high-performing month, showing the highest cumulative 3-month retention for the year.

Repeat Purchase Depth 2024



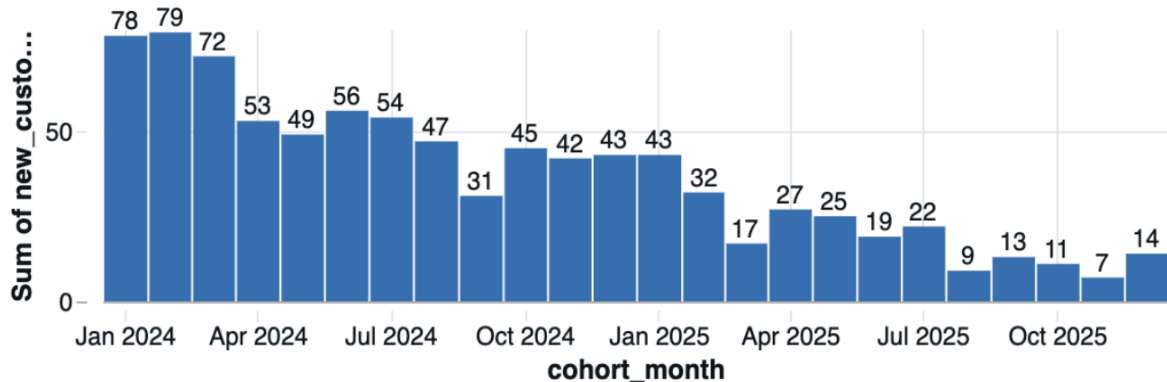
Referencing: "Repeat Purchase Depth in Percentage 2024" Chart

- Repeat Depth Trend:** Loyalty for a 2nd purchase (`repeat_2plus`) was strong in early 2024 (near 50%). Deep loyalty (4+ purchases) remained consistent but at a lower level of approximately 10-15%.

2. Full Period Analysis (2024 - 2025 Combined)

Cohort Size (Total Trend)

Total Customers (New Customers) 2024 - 2025

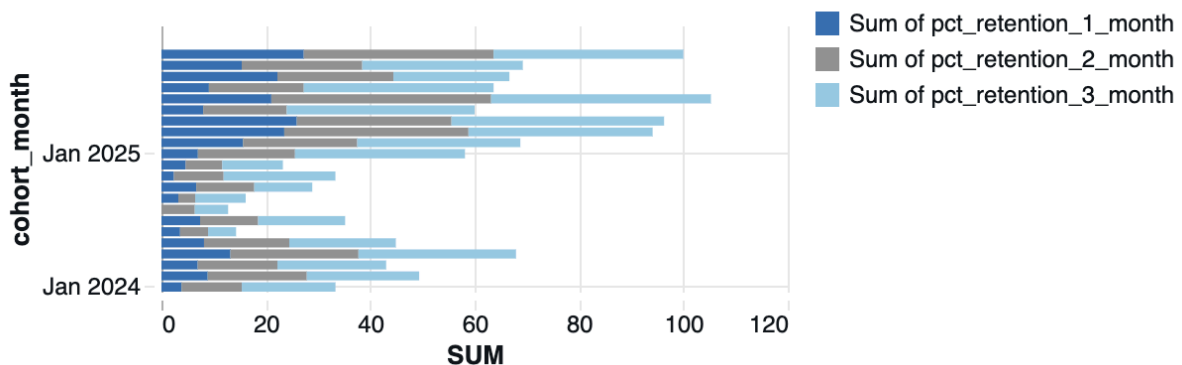


Referencing: "Total Customers (New Customers) 2024 - 2025" Chart

- **Trend:** The downward trend in new customer acquisition continued into 2025. The smallest cohorts are observed toward the end of 2025, reaching a minimum of 7 new customers in November 2025.

Retention to 2nd Purchase (Total Trend)

Cohort Segmentation in Percentage 2024 - 2025

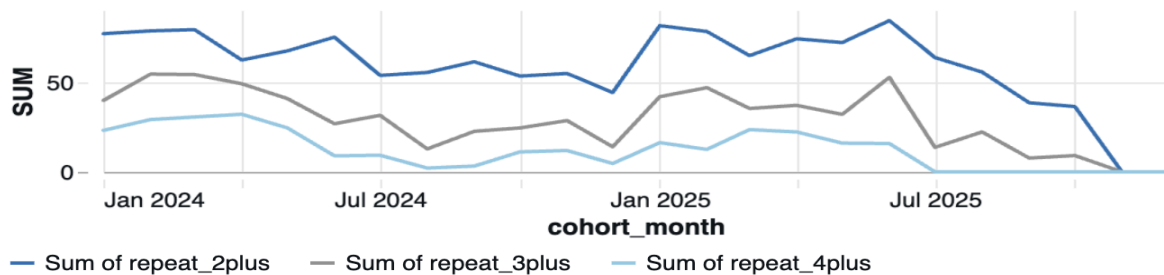


Referencing: "Cohort Segmentation in Percentage 2024 - 2025" Chart

- **Trend:** Despite smaller cohort sizes, the relative retention quality in early 2025 (Jan–Apr) and a peak performance in June and October appears significantly higher than the 2024 average. This suggests that while fewer customers are being acquired, they are of higher "quality" or loyalty.

Repeat Purchase Depth (Total Trend)

Repeat Purchase Depth in Percentage in 2024 - 2025



Referencing: "Repeat Purchase Depth in Percentage 2024 - 2025" Chart

- **Trend:** The first half of 2025 (Jan–Jun) showed the strongest results for repeat purchase depth. This is evidenced by high percentages in the **2+, 3+, and 4+ purchase categories**, indicating that customers acquired in early 2025 were exceptionally loyal. In August 2025, we observe a 2nd purchase rate of 55.56%. However, the deeper metrics (3+ and 4+ purchases) drop significantly toward the end of the timeline.

3. Critical Data Observations & Analysis

- **Delayed Repeat Purchase Effect:** * Customers generally tend to make repeat purchases later in their lifecycle.
 - This explains why **repeat_3plus** and **repeat_4plus** metrics drop to 0% for the most recent cohorts (late 2025). This is not a sign of poor performance but a result of **Cohort Maturity**: these customers have not yet had enough time to reach their 3rd or 4th order.
- **Missing Data Point (Dec 31, 2024):**
 - There is a verified gap in the source data for **December 31, 2024**.
 - While this is a single day, it should be noted that December 2024 totals may be slightly understated, potentially affecting the transition metrics into January 2025.

4. Executive Summary

- **Data Quality:** The 2024/12/31 is missing in the original Dataset ecom_orders_2024_rebalanced.csv. Thus we have to prove it if there are data for this date as the data are not complete for the full year.
- **Acquisition:** Significant decline over 24 months. Focus should be placed on why top-of-funnel acquisition is shrinking.
- **Retention:** 1–3 month retention is volatile but shows improvement in early 2025 cohorts.
- **Repeat Depth:** The 2+ purchase rate is healthy (often >50%), but the 4+ purchase "power user" segment is narrowing.
- **Final Note:** Recent cohorts (Q3/Q4 2025) should be re-evaluated in 6 months to allow for natural purchase cycles to complete.

Single-Purchase Analysis

- **Isolation of "One-Time Buyers":** It is recommended to analyze customers who have only made one purchase separately to calculate the exact churn rate immediately following initial acquisition.
- **Correlation with Acquisition Decline:** Given that new customer acquisition hit a low of 31 in September 2024 and dropped further to 7 in November 2025, identifying the percentage of non-returning customers is vital to understanding the return on investment for marketing.
- **Quality Validation:** While early 2025 cohorts show higher loyalty (2+ purchases often >50%), a single-purchase analysis would clarify if this is due to a genuine increase in customer quality or simply a shrinking pool of "low-quality" buyers.
- **Predictive Indicator for Recent Cohorts:** For the Q3 and Q4 2025 cohorts, where deeper metrics like 3+ and 4+ purchases currently drop to 0% due to "Cohort Maturity", tracking one-time buyers helps distinguish between customers who *will* buy again and those who have already churned.

Strategic Recommendations & Next Steps

1. Data Integrity & Pipeline Audit (Technical)

- **Missing Data Verification:** A critical audit of the **Bronze-to-Silver layer transition** is required to verify the data for **December 31, 2024**. We must confirm if the missing records on New Year's Eve are a source system issue or a transformation anomaly.
- **Validation of Year-End Transition:** Ensure that the "First Purchase Date" logic in the Silver Layer correctly handles orders from the final day of the year to prevent skewed results for the January 2025 cohort.

2. Pivot to Retention-First Strategy (Business)

- **Maximize "Power User" Value:** Since the 2025 cohorts show higher relative loyalty (up to 55.56% 2nd purchase rate), the business should shift budget from failing broad acquisition to dedicated loyalty programs for these "quality" customers.
- **Single-Purchase Recovery:** Launch a re-activation campaign targeting the large pool of "One-Time Buyers" from 2024. It is significantly more cost-effective to win back these users than to acquire new leads at the current 2025 volume (only 7–12 customers/month).

3. Performance & Maturity Monitoring (Analytical)

- **Niche-Market Analysis:** Investigate if the 91% drop in acquisition volume is a result of a deliberate shift toward higher-quality, niche audiences or a failure in marketing reach.
 - **Cohort Maturity Buffer:** Postpone final budget decisions for Q3/Q4 2025 cohorts for 6 months. Due to the natural purchase cycle, their "Deep Loyalty" (3+ purchases) metrics are currently understated and need time to mature.
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